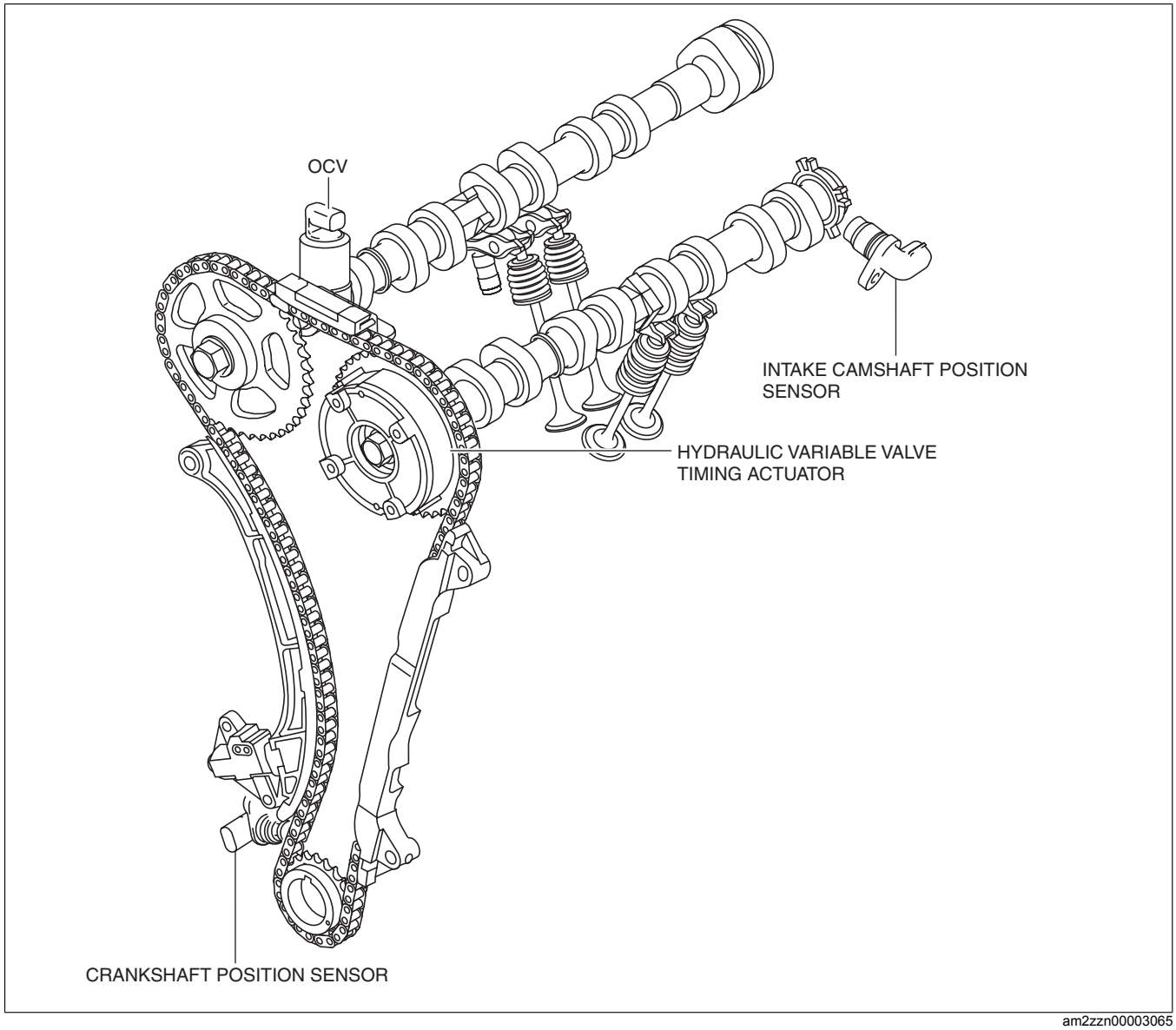
VARIABLE VALVE TIMING MECHANISM [SKYACTIV-G 1.3, SKYACTIV-G 1.5 (WITH 4-1 EXHAUST SYSTEM)]

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Outline

- The variable valve timing mechanism changes the valve timing.
- By optimizing the change in the valve timing, the internal EGR amount and effective compression ratio are controlled, pumping loss is reduced, and the combustion is controlled optimally.
- A hydraulic pressure type variable valve timing mechanism has been adopted on the intake side.

Structural View



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Structure

Part name	Function
Hydraulic variable valve timing actuator	(See HYDRAULIC VARIABLE VALVE TIMING ACTUATOR [SKYACTIV-G 1.3, SKYACTIV-G 1.5 (WITH 4-1 EXHAUST SYSTEM)].)
OCV	(See OIL CONTROL VALVE (OCV) [SKYACTIV-G 1.3, SKYACTIV-G 1.5 (WITH 4-1 EXHAUST SYSTEM)].)
Intake camshaft position sensor	(See CAMSHAFT POSITION (CMP) SENSOR [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
Crankshaft position sensor	(See CRANKSHAFT POSITION (CKP) SENSOR [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)

Operation

- The variable valve timing mechanism has the following four modes:

- The four modes are determined by the operation of the OCV and hydraulic variable valve timing actuator.

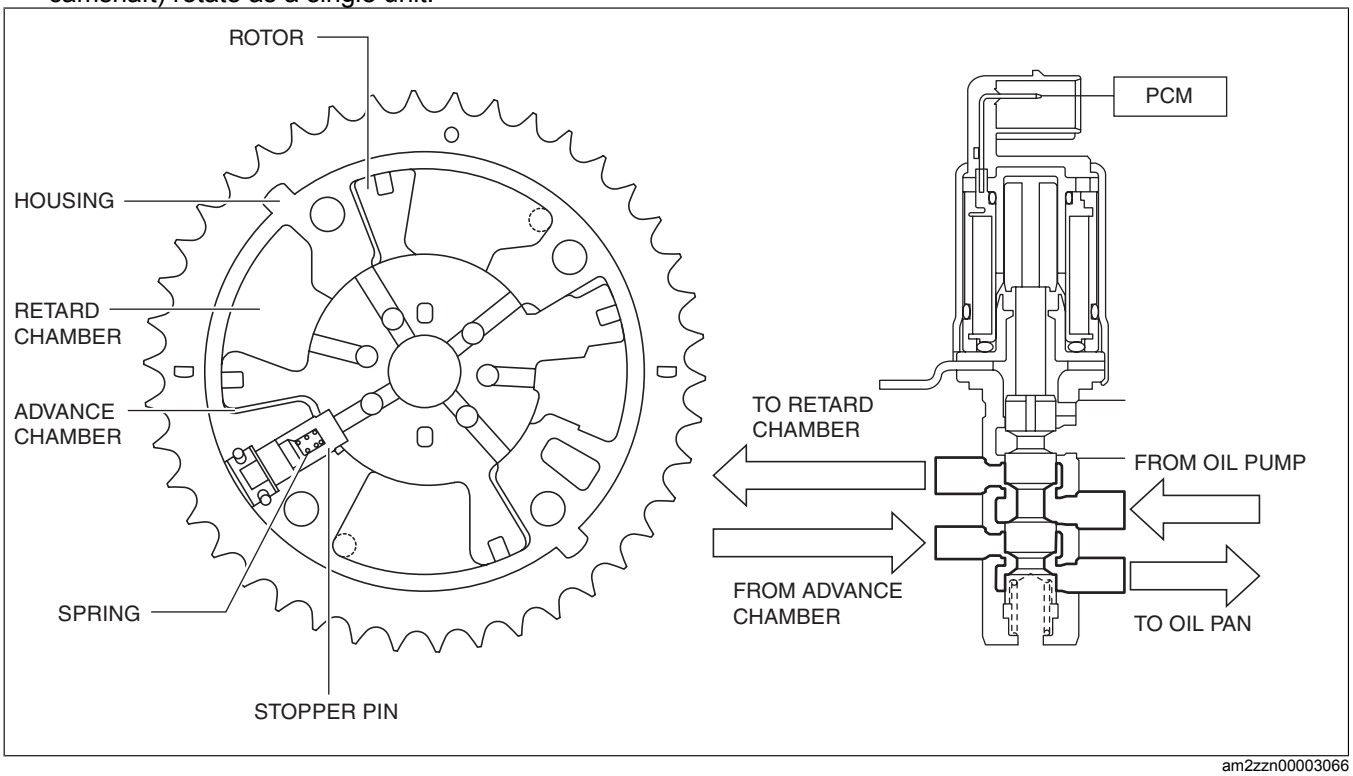
Mode	OCV operation (spool valve)	Hydraulic pressure	Hydraulic variable valve timing actuator operation		Valve timing
			Stopper pin	Rotor	
Fixed mode	Maximum retard position	To retard chamber	×	Maximum retard position	Maximum retard
Advance mode	To advance side	To advance chamber	—	Move to advance side	Advance
Retard mode	To retard side	To retard chamber	—	Move to retard side	Retard
Hold mode	Middle point	To advance/retard chamber	—	Hold at optimum position	Hold at optimum position

× : Engages with rotor depression

— : Disengages from rotor depression

Fixed mode

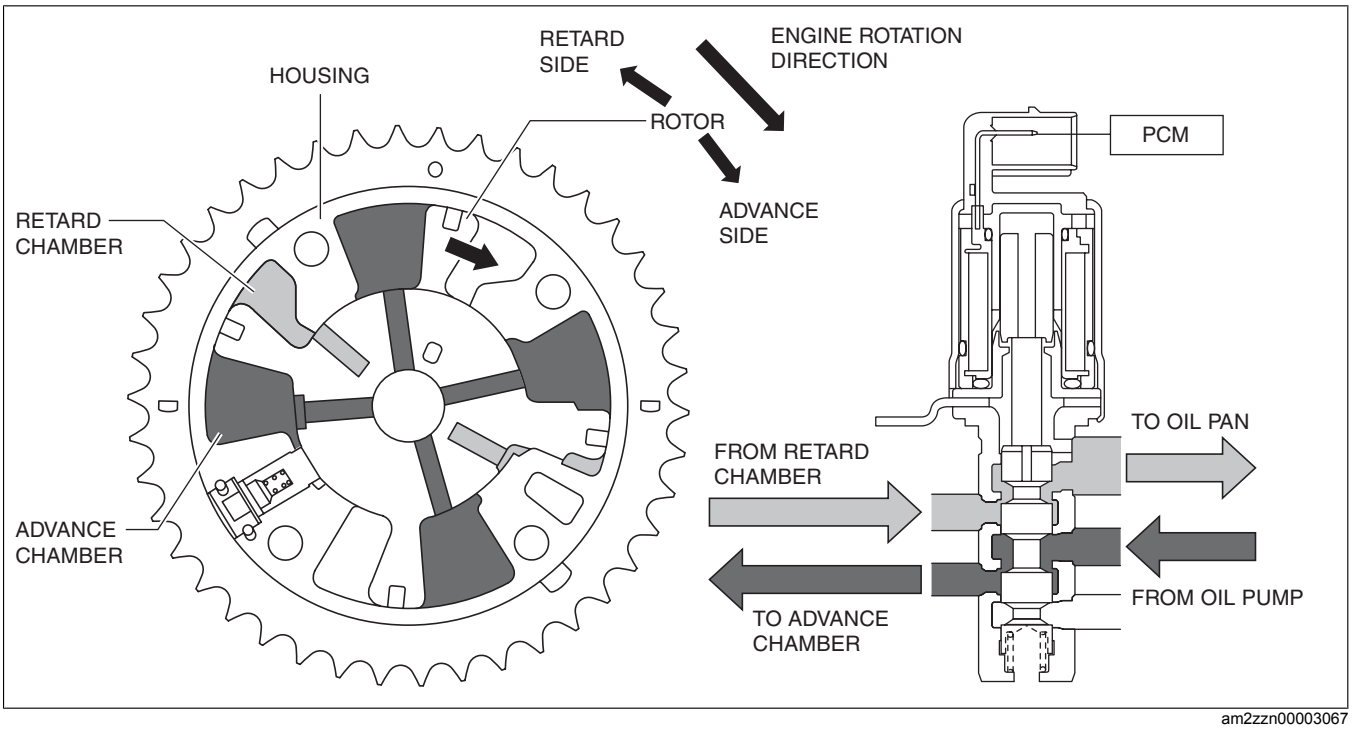
- The rotor depression at the maximum retard position and the stopper pin in the hydraulic variable valve timing actuator are engaged by the spring force. Due to this, the housing (intake camshaft sprocket) and the rotor (intake camshaft) rotate as a single unit.



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Advance mode

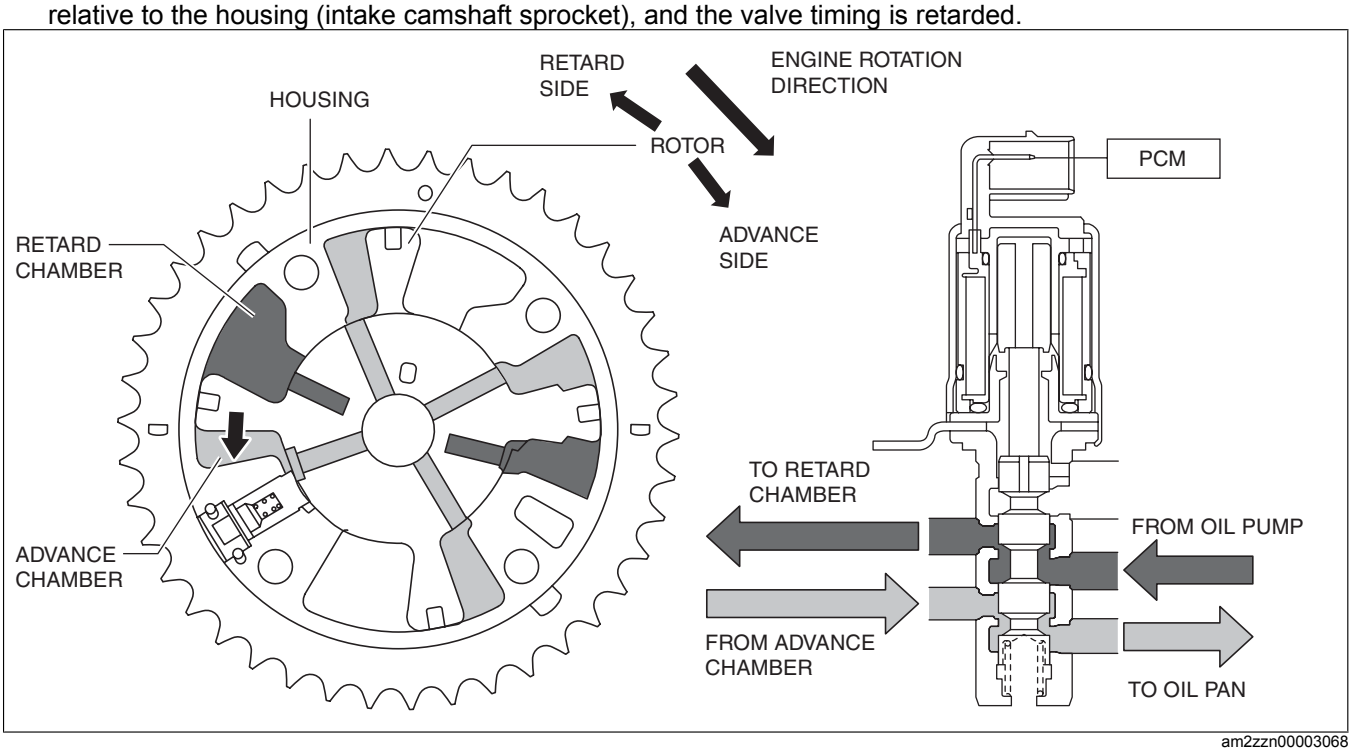
- The spool valve in the OCV moves to the advance side and hydraulic pressure is applied to the advance chamber in the hydraulic variable valve timing actuator by a signal from the PCM. Due to this, the rotor (intake camshaft) rotates in the advance direction relative to the housing (intake camshaft sprocket), and the valve timing is advanced.



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Retard mode

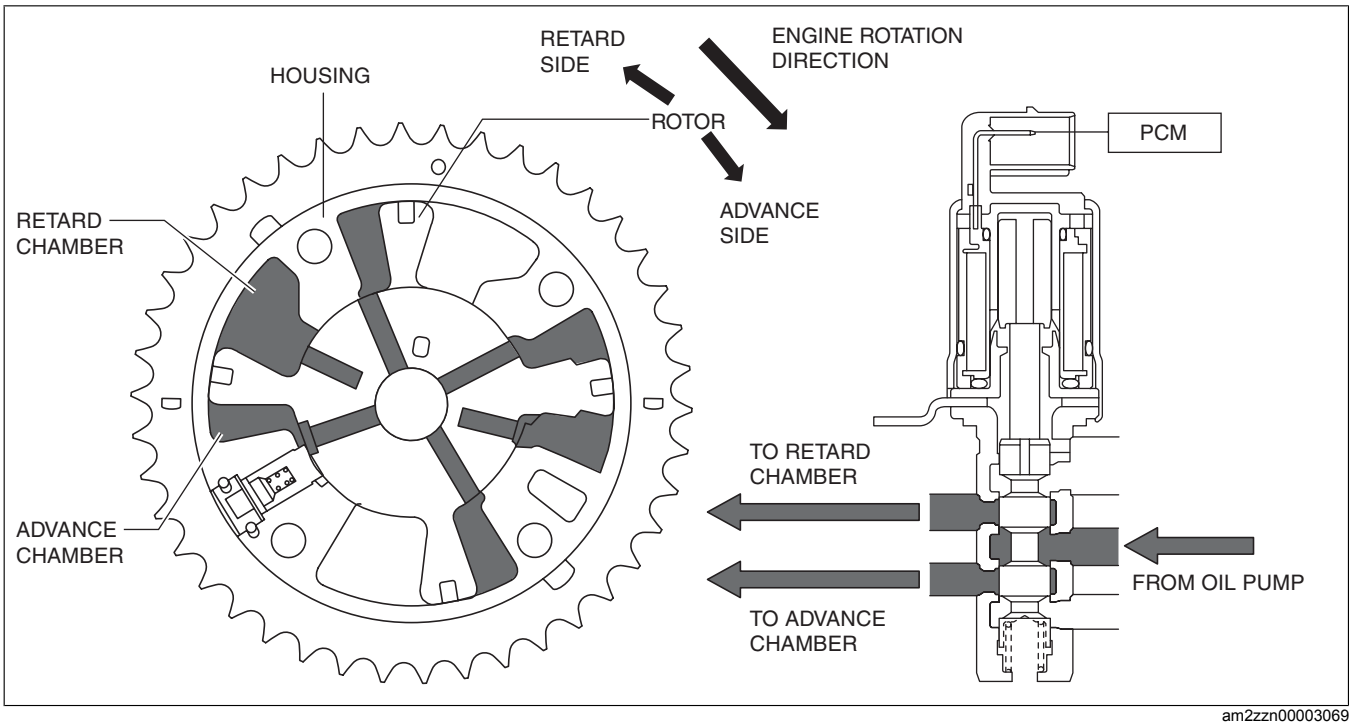
- The spool valve in the OCV moves to the retard side and hydraulic pressure from the oil pump is applied to the retard chamber by a signal from the PCM. Due to this, the rotor (intake camshaft) rotates in the retard direction relative to the housing (intake camshaft sprocket), and the valve timing is retarded.



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Hold mode

- The spool valve in the OCV is positioned the area around the middle point at advance and retard by a signal from the PCM. At this time, hydraulic pressure is applied to both of the advance and retard chambers in the variable valve timing actuator. Due to this, there is no change in the relative angle of the rotor (intake camshaft) and the housing (intake camshaft sprocket) and the current valve timing is held.



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Note

- To switch to the fixed mode (lock-pin in hydraulic variable valve timing actuator engaged with rotor depression) when the ignition is switched OFF (LOCK), the engine stop condition is extended (within approx. 0.5 s) under the following conditions: Due to this, engine re-startability has been improved. (ATX)
  - — During or right after fast idle increase (during control of catalytic converter earlier activation)
  - — Right after operating to D/R/M position while the vehicle is parked at high-altitude, or operating from D/R/M position to P/N position